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Internet of Things for gaming: A review

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ABSTRACT

The Internet of Things (IoT) is an advanced tech that transfers game data such as graphics, text, and other data over real-time networks. Game player surroundings can be sensed, or movement or situation can be observed for wearable devices that use sensors and actuators, this increases sales of wearables from 18 million in 2015 to 533.6 million units by 2021. The information of game players gathered by sensors and actuators is sent for further action in the game engine. In this paper, we survey and analyze the involvement of IoT technology sensors and wearables in serious and social gaming. Further, we reviewed IoT applications for gaming, as well as their challenges and limitations, with open research issues for future development.

1. Introduction

The Internet of Things refers to a technical system that connects various devices to the Internet to realize information sharing and automatic management among devices [1]. Since its development in the 1990 s, the Internet of Things has become a hot topic in the global technology industry [2]. The gaming industry is also growing day by day and provides services to game users via the online cloud to reduce the cost of purchasing hardware [3]. The advancement and development of technology, such as virtual reality headsets, increase the gaming experience, attract more people to play games, and increase the gaming industry's revenue [4]. The addition of IoT devices, such as wearable sensors and others, increases game players' experience and excitement [5,6]. It allows users to connect worldwide with lenses, experience more, and feel more.

The integration of new technologies such as cloud computing, virtual reality, and IoT has changed the concept of old conventional gaming, where high-performance computing and paid game emulators are required to install and play games [7]. In addition, cloud computing models reduce the hardware cost of gaming, which requires new emulators of fun, and also provide an opportunity for game players to play various games online on a pay-per-play basis and connect with multiplayer around the world [8]. Virtual reality headsets increase graphics quality and focus users on rounds for more excitement and engagement with games [9]. IoT gaming simulators provide a natural feel to game players that they are not playing a game but doing things related to gaming, such as car racing car, driving in real-time, and in a natural environment, and the game of cricket is like playing cricket on the natural ground with other players [10,11]. Fig. 1 MYO armband allows game players to control the game with hand gestures, and Fig. 2 shows illustrations of game players using a VR box for gaming and controlling gaming with the hand control device [12].

The paper aims to provide the state-of-the-art role of IoT in modern game development and play. Further, this research provides IoT architecture, IoT in the gaming industry, IoT Wearables for Social Gaming, IoT Wearables and Sensors for Serious Gaming, IoT applications for gaming, and challenges and limitations of previous research work.

2. Methodology

This study depends upon the systematic reviews of the IoT for gaming, limitations, and open research issues. This is the recent study of the IoT used in gaming and its technical limitations during the adoption of IoT. Furthermore, Systematic Literature Review (SLR) and System Mapping Study (SMS) have introduced IoT for gaming. The subsections of SLR have been discussed in detail.

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1875-9521/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

2.1. Research questions

RQ.1. What sensor technologies are commonly used in enabled serious and social gaming environments? How do they contribute to improving the user experience and gameplay?

Ans: Motion sensors are frequently used in both social gaming environments. These sensors greatly enhance the user experience by allowing gameplay through motion tracking. Additionally, they provide feedback based on data, further enhancing the overall gaming experience. Furthermore, discussed in sections 3 and 4.

RQ.2. How do wearable devices, with sensors and actuators, enable real-time data collection and interaction in gaming scenarios to immerse players and increase engagement?

Ans: During gameplay, wearable devices equipped with sensors and actuators allow for the real-time collection of data by capturing the movement information of players. Moreover, as discussed in Section 5.

RQ.3. What are the significant applications of technology in gaming, and how do they affect gameplay dynamics, social interactions, and user behavior?

Ans: Technology has impacted gaming, with applications like augmented reality, virtual reality, and cloud gaming, leading the way. These innovations have transformed gameplay by creating experiences that enable multiplayer interactions and providing ways for gamers to engage and collaborate within their communities. Moreover, discussed in section 5 also.

RQ.4. What challenges and limitations arise when integrating sensors and wearables into gaming environments, and how can we effectively address these obstacles?

Ans: There are challenges and limitations when incorporating sensors and wearables into gaming environments. These include compatibility issues, privacy concerns, and the possibility of glitches. They are further discussed in section 8.

RQ.5. What are the current trends and emerging developments in IoT-based gaming technologies, and how are they shaping the landscape of serious games and social gaming experiences?

Ans: The latest developments in gaming technology for the IoT involve incorporating devices, like smartphones and smartwatches, to enhance gameplay with reality and location-based functionalities. Detailed answers are available in section 9.

Selection criteria and Non-Selection criteria

Our selection criteria are based on scientific papers published in well-known journals and reputed research agencies. Those related papers are in standard language "English" and extracted from well-known databases like "IEEE," "Elsevier," "Springer," "Emerald," "Wiley Online Library," "MDPI," and "Google Scholar." These are related to the 2012 to 2024 years based on this SLR.

Table 1 shows the selection and non-selection criteria for the published papers.

2.2. Searching key terms techniques

The most important keywords were used during the search ("IoT for Gaming," "Social Gaming," "Serious Gaming," "IoT Applications," "IoT Architecture," "Wearable Sensors for Gaming," and "Sensing devices for Serious Games") in well-known journals and google scholar database.

SLR steps

This SLR and SMS entirely focused on well-known and recently published papers:

1. The Used papers inside this SLR were extracted from above databases.
2. The critical research questions have been highlighted and answered.
3. Nonrelated and unsatisfied work has been excluded from the SLR.
4. Most collected data relate to published work from their titles, abstracts, and well-known journal databases.
5. Finally, the most relevant data and published papers are cited in this paper to increase the effectiveness of our SLR.

This paper is organized into 10 sections; Section 3 provides details of IoT and its architecture, and Section 4 is based on how IoT will play its role in the gaming industry. Similarly, in section 5, we discussed IoT wearables for social gaming, and section 6 is based on how IoT devices and sensors play a role in serious games. Section 7 presents details of IoT gaming applications, and section 8 is based on the challenges and limitations. Section 9 is based on the open research, and finally, in section 10, we conclude our work.

3. Internet of things

Intent IoT is a new type of technology that allows users to avoid the need to pay attention to the internal technical implementation details of the IoT network [13]. They only need to use natural conversational language to express network application requirements, thereby achieving automation, high reliability, and closed-loop real-time verification. The network service has changed the traditional Internet of Things network construction and management methods and improved

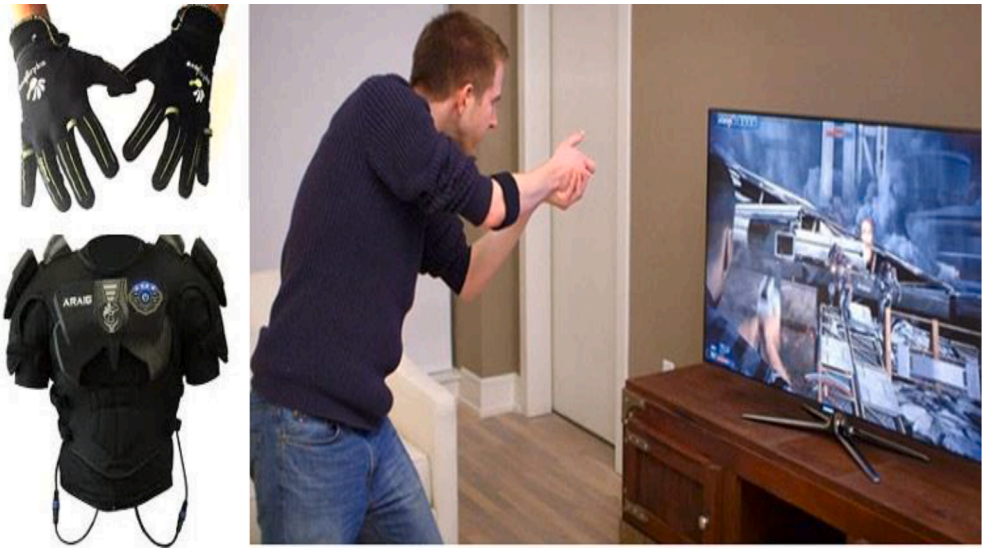


Fig. 1. MYO armband game simulator.

the user experience. At the same time, being able to quickly forward and process the huge information data traffic in the networks is the future development trend of the IoT [14,15].

With the rapid development of emerging technologies such as artificial intelligence, cloud computing, and big data, the IoT, as an important link connecting all things, has also developed rapidly in recent years [16,17]. The development of the IoT has brought new opportunities and challenges to all walks of life. In the future, the IoT will bring more changes and progress to society.

First, the IoT provides new business opportunities for enterprises [18,19]. Through the networking of equipment and sensors, enterprises can realize remote monitoring and control of equipment, greatly reduce management costs, and improve production efficiency. For example, in the industrial field, the application of the IoT can help enterprises realize the fault prediction and maintenance of equipment, thereby improving equipment utilization and production efficiency.

Secondly, the IoT provides new opportunities for the development of smart homes, smart cities, and other fields [20]. In the smart home, by connecting home appliances, furniture, and other equipment to the Internet, the automatic control and intelligent operation of the house can be realized, and the comfort and convenience of life can be improved. In a smart city, by connecting information such as transportation and public facilities to the Internet, the intelligent management and operation of the city can be realized, and the operating efficiency of the town and the quality of life of residents can be improved [21].

Finally, IoT also brings new safety and security mechanisms to society [22]. For example, in the field of smart medical care, by connecting medical equipment to the Internet, medical resources can be shared and optimized, and the quality and efficiency of medical services can be improved. In the field of intelligent transportation, by connecting transportation facilities with the Internet, real-time monitoring and regulation of traffic information can be realized, and traffic safety and convenience can be improved [23].

The development of the IoT has brought new opportunities and challenges to society and, at the same time, brought us new thinking and exploration. With the continuous innovation and development of the IoT technology, the Internet of Things will bring us a broader future.

3.1. IoT architecture

1) Application layer

Provide support for upper-layer applications according to user needs, and network managers can implement network control and management functions through programmable interfaces [24].

Table 1
The criteria of selection and non-selection for the published papers.

Selection Criteria	Non-Selection Criteria
When selecting the published papers, we focused on "IoT for Gaming," "Social Gaming," "Serious Gaming," and "IoT Applications." The data has been extracted from past and recent published papers, such as from 2012 to 2024. In a deep understanding, we have downloaded more than 500 published papers during filtering process, we have selected 20 % IEEE, 30 % Springer, 10 % Willey, 8 % Elsevier, 5 % CRC, 5 % MDPI, 5 % Emerald, 7 % other Scopus journals. Total we have selected 117 papers. These papers are related to our SLR.	Our selection criteria are focused on authentic data. Due to the strict selection criteria process, we have excluded the "blogs," "websites," "Wikipedia," and "technical reports." Because such type of above-related source data is not verified and authenticated for this SLR, we have excluded the many research papers because of not published in well-known journals of publishers.

2) Network layer

An important component of the Network-enabling layer is the intent-processing module [25]. This module includes intent translation and verification and is mainly responsible for processing user intent from Intent NBI. Basic functional modules mainly include topology calculation, path calculation, routing and forwarding, etc. The positioning of IoT devices can be achieved through IP positioning or name positioning. Therefore, an information name library is defined to store device name information. IoT device management realizes functions such as access control and overall perception of IoT devices. The basic support module includes five parts: module management, event system, log system, time management, and storage service, which provide overall support for the operation of other modules [26].

The Network enabling layer provides Intent NBI for the application layer [27]. Through this interface, users can use normal conversational language to describe their needs for network applications. The controller is regarded as an intelligent black box integrating core network services to build network applications. Meet user requests; centrally manage and control all network infrastructure equipment by calling the southbound interface and through the OpenFlow protocol. At the same time, the intent-enabling layer can dynamically add, delete, and modify the flow table entries of the OpenFlow switch through the management of the OpenFlow switch, thereby changing the forwarding rules of the OpenFlow switch and achieving centralized control of the underlying devices.



Fig. 2. Game player with virtual reality head set.

3) Data Processing layer

The data Processing layer is composed of various switching devices in the network [28]. Its main function is to support OpenFlow switches and access networks, core networks, and other network layers. This layer implements high reliability and security through the IoT gateway and the lower sensing layer equipment. The IoT gateway mainly includes a communication module, a central control unit, and a data collection module. It has certain data processing and equipment management capabilities. It can perform coarse-grained processing of data from the bottom layer and then report it to the upper controller improving the efficiency of the controller [29].

4) Information Sensing layer

The main function of the information sensing layer is to collect underlying data information through smart terminals with various sensors [30]. The collected information communicates with the upper layer through the IoT gateway to provide data information for the business application layer to execute various commands.

5) Intent NBI

Intent NBI is located between the application layer and the Network layer [31]. Network applications use Intent NBI to communicate with the controller to express network behavior and define configuration requirements and program forwarding requirements. Intent NBI hides the underlying details of network objects and services, allowing users to express intentions in a declarative rather than imperative manner.

6) Southbound interface

The southbound interface of IoT is used to define the communication protocol between the controller and network devices. OpenFlow is a mature southbound interface guidance forwarding plug-in that SDN controllers use to update and insert flow rules that specify the relevant actions that need to be performed for each flow through a given network device. The IoT architecture is shown in Fig. 3.

3.2. IoT challenges

As one of the most popular technologies at present, the IoT has continuously expanded its application fields and has a very broad market prospect. However, with the continuous development of IoT technology, it also faces a series of challenges.

First of all, the IoT involves a large amount of data processing and transmission, and security issues have become a major problem in development [32]. With the increase in the number of IoT devices, security issues such as network attacks and data leakage are also increasing. Ensuring the security and privacy protection of IoT devices

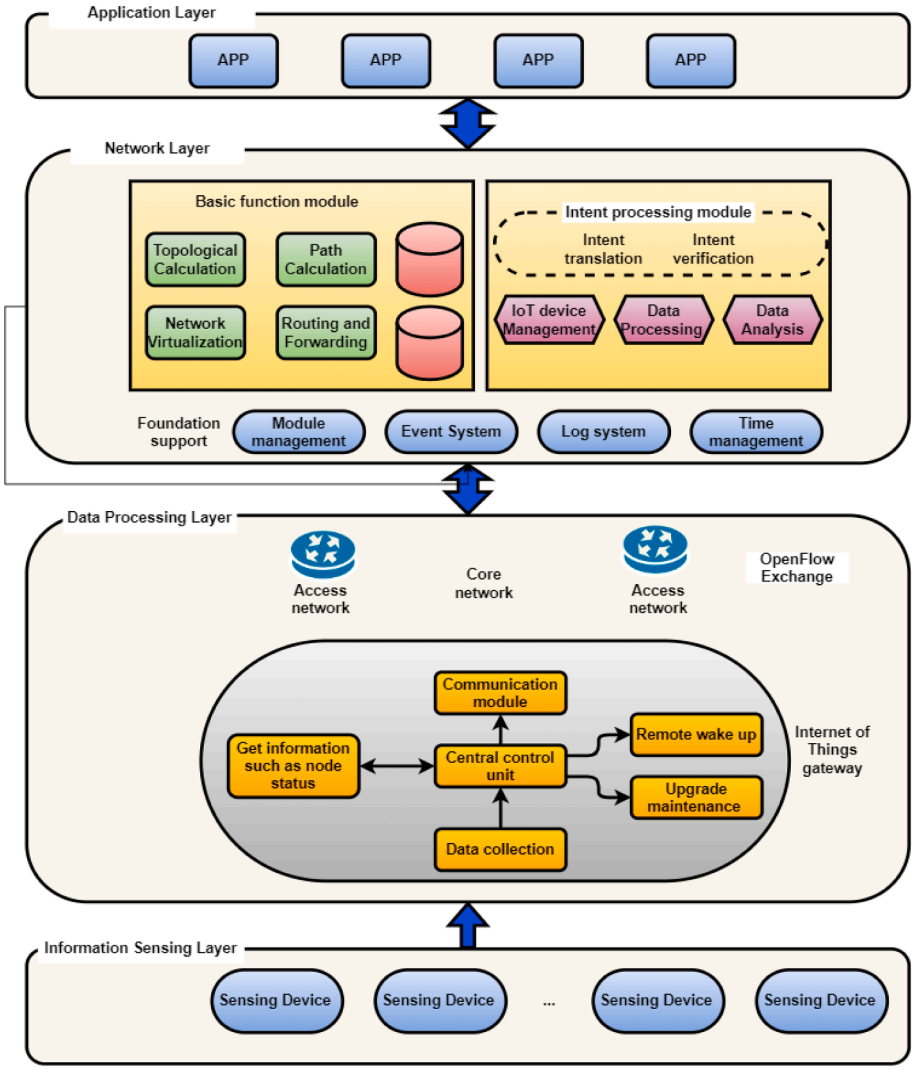


Fig. 3. IoT architecture.

and networks has become a problem that must be solved in the development of the IoT.

Secondly, the widespread application of the IoT has also brought about the difficulty of technology diversification and standardization [33]. The IoT involves many different types of devices and technologies, and how to work together and unify standards is an important issue [34]. At present, the IoT devices of various manufacturers need to follow a unified standard, resulting in compatibility problems between devices, which brings certain difficulties to the application of the IoT.

In addition, the energy consumption of IoT devices is also worthy of attention [35]. Since IoT devices need to work stably for a long time, efficient energy consumption control technology is required. At present, many IoT devices have short battery life and frequent charging cycles. Therefore, how to improve the energy efficiency and lifespan of devices has become a difficult problem in the development of IoT.

4. Internet of Things in the gaming industry

The purpose is entertainment and interaction, covering a wide range of game types, including electronic games, mobile games, online games, and more. With the continuous advancement of technology, the game industry has gradually incorporated many new technologies, such as virtual reality, augmented reality, artificial intelligence, etc., to make the game experience more colorful [36,37].

In the gaming industry, IoT technology is also beginning to be applied. Game developers can use IoT technology to integrate the real world with the virtual world to provide players with a richer gaming experience [38]. For example, players can input their real-time physiological data into the game by wearing a smart bracelet so that the game can adjust the difficulty and expertise based on these data. In addition, IoT technology can also be used to improve the logistics and inventory management of the game and increase its production and operational efficiency.

4.1. The development of Internet of Things in the gaming industry

The combination of the game industry and the IoT can be traced back to nearly ten years ago [39]. With the continuous development and popularization of the IoT technology, the game industry has gradually begun to apply it.

In 2021, the global sales of smart wearable devices reached 81 million units, many of which are used to interact with games, such as Fitbit and Apple Watch [40]. In 2016, the launch of the Pokémon Go game ignited a global boom [41]. The game uses IoT technology to combine virtual reality with the real world. Users need to interact with virtual reality through the GPS positioning of mobile devices. In 2017, Microsoft released the Xbox One X game console, which has a built-in Cortana smart voice assistant, allowing users to control games through voice commands [42]. In 2018, Amazon released the Alexa smart speaker, which can interact with games, such as launch games or control operations in games through voice commands [43]. In 2019, Huawei released its smart speaker HiAI, which can interact with games, such as launching games or controlling in-game operations through voice commands [44]. In 2020, the game company Epic Games launched the Unreal Engine 5 game engine, which can be connected to IoT devices to achieve a more intelligent game experience [45]. In 2021, with the popularization of 5G technology, game companies will begin to combine IoT technology with 5G technology to launch a more intelligent and high-definition gaming experience [46]. Sales of IoT smart wearables are given in Fig. 4.

4.2. Internet of things technology for game industry

Smart gaming device: The Internet of Things technology can intelligent game devices such as game machines, controllers, and head-mounted displays [47]. For example, Microsoft's Xbox One game console has built-in sensors, such as temperature sensors and motion sensors, which can be connected with other devices through the Internet of Things technology, allowing users to obtain a more intelligent and convenient gaming experience [48,49]. In addition, some manufacturers have also launched smart game controllers, such as the Steam controller and Xbox Elite controller, which have built-in sensors and feedback mechanisms to help users better control the game.

Networked game platform: IoT technology can also help gaming companies develop smarter, connected gaming platforms. For example, Sony's PlayStation Network and Microsoft's Xbox Live are networked gaming platforms that allow players to play online games with other players, download game content, watch game videos, etc., anytime, anywhere [50]. In addition, some game companies have also developed smart game-supporting platforms using IoT technology, such as EA's

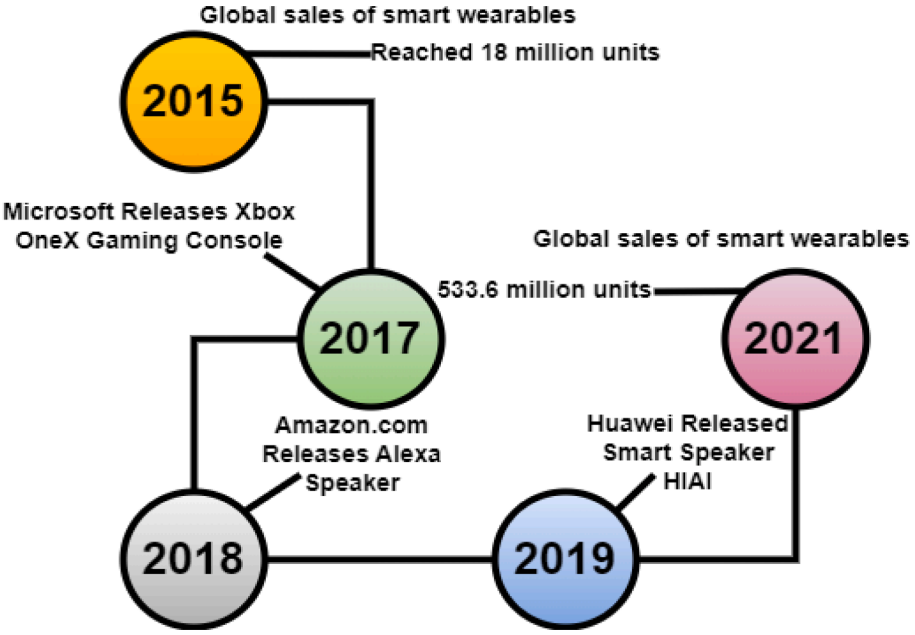


Fig. 4. Sales of IoT smart wearables.

Origin platform and Blizzard's Battle.net platform, which can help users manage their game libraries, social relationships, and game data [51].

Smart game venues: IoT technology can also help game companies create more intelligent and technologically perception game venues. For example, Tencent Game World is a smart game venue that has built-in a large number of sensors and smart devices, allowing users to obtain a more realistic and immersive experience in the game [52]. In addition, some game companies have also used IoT technology to create virtual reality gaming venues, such as HTC's Viveport Arcade and Oculus VR Arcade, which allow users to play games, socialize, and interact in a virtual reality environment [53].

Data analysis and artificial intelligence: IoT technology can also help game companies conduct more accurate and intelligent data analysis. For example, by collecting game data, behavioral data, and social data of players, game companies can analyze the game behavior of players [54].

4.3. The future of IoT in the gaming industry

More connected devices: In the future, the game industry will make more use of the Internet of Things technology to realize the interconnection of various devices. For example, players can interact with the game through devices such as smartphones, tablets, smartwatches, and smart glasses [55]. Game producers will also apply IoT technology to game hardware, such as VR devices, handles, and head-mounted displays [56].

Smarter gaming experience: IoT technology can help game producers better understand players' preferences and behavioral habits so as to provide players with a more intelligent gaming experience [57]. For example, games can adjust difficulty, characters, and items based on the player's individual preferences.

More realistic game environment: With the development of IoT technology, the game industry can more realistically simulate various environments in the real world, including sound, temperature, smell, light, etc [58]. This will lead to a more realistic gaming experience, enhancing the player's sense of immersion and engagement.

More social interaction: IoT technology can make games more social, and players can interact more intelligently and richly with other players, such as sharing game progress, achievements, and gameplay through social media [59]. In addition, game producers can also use IoT technology to enhance the interaction between players, such as providing online chat, real-time voice, and other functions.

5. IoT wearables for social gaming

5.1. Social games

Social gaming is an online interactive game played through social networking platforms [60]. Such games typically allow players to interact, communicate, and cooperate with other players in the game. Social games can be played in a variety of ways; some are role-playing-based, some are competitive and strategy-based, and some are puzzle and exploration-themed.

The biggest feature of social games is their social nature. Players can make new friends with other players in the game, share game experiences, and challenge game levels together. Social games also have relatively simple rules and operations and do not need to spend too much time and energy learning the rules of the game, so they are very suitable for those who want to relax and have fun in their spare time.

In addition to entertainment, social games can also help players improve some social skills, such as communication skills, collaboration skills, teamwork, etc. Cooperating with other players in the game to complete tasks and overcome difficulties can make players feel the joy and sense of accomplishment of working together [61].

Social games have become more and more popular in modern society, especially the popularity of mobile devices, which makes such games more convenient and easier to use [62]. In the future, social games will continue to develop, providing people with more colorful game experiences and social communication platforms.

5.2. Social games based on IoT technology

Social games based on IoT technology refer to the use of IoT technology to create a richer and more realistic game experience, allowing players to control virtual characters through IoT devices or use IoT sensors to capture real-world information in the game. Interact with other players.

Pokémon Go: This is a game developed by Niantic that uses IoT technology to bring together the real and virtual worlds. Players need to use the smartphone's GPS function and camera to find and capture virtual elves in the real world [63].

VRChat: This is a virtual reality social platform game. Players can use a virtual reality head-mounted display (VR helmet) to enter the virtual world in the game to communicate and interact with other players [64].

Ingress: This is a game developed by Niantic, similar to Pokémon Go, where players need to use the GPS function and camera of their smartphones to collect power and control virtual doors in the real world [65].

Sky Children of the Light: This is a game developed by That game company that allows players to use a smartphone or tablet to enter a virtual world within the game to explore and interact with other players [66]. The game uses IoT technology to perceive the player's gestures and movements so that the player can control the virtual character in the game through gestures.

5.3. IoT wearables for social gaming

With the development of IoT technology, more and more wearable devices are being designed and developed to support various social games and interactive experiences. Fig 5. illustrates IoT wearable devices and details also described in subsections.

VR Head-Mounted Display: A virtual reality head-mounted display is a device that immerses players in a virtual gaming world and can provide an amazing social gaming experience. Players can interact with other people in the virtual world and share the gaming experience [67].

Smart bracelets: Smart bracelets can monitor the user's movement and physical condition and interact with other players [68]. For example, in a team game, smart wristbands can track each player's activity level and send reminders and encouragement messages to players to encourage them to continue participating in the game.

Smart Watch: A smartwatch can communicate with other devices and applications and provide real-time game status and notifications. For example, in a social game, a smartwatch can send reminders and messages to players to encourage them to interact with other players [69].

: Smart glasses can provide augmented reality and virtual reality functions to help players better integrate into the gaming experience [70]. For example, in a team game, smart glasses can provide players with information about the game scene and player status.

Smart Gloves: Smart gloves can track the player's hand movements and simulate hand movements in the game to enhance the realism of the game. For example, in virtual reality games, smart gloves can simulate the player's hand movements, allowing players to experience the game world better [71].

Smart shoes: Smart shoes can track a player's steps, movement speed, and direction and share this information with other players [72]. For example, in an urban exploration game, smart shoes could track a player's location and send tips and quest information to other players to encourage them to stay engaged.

Smart headsets: Smart headsets can track players' voices and communicate with other players in real-time [73]. For example, in team

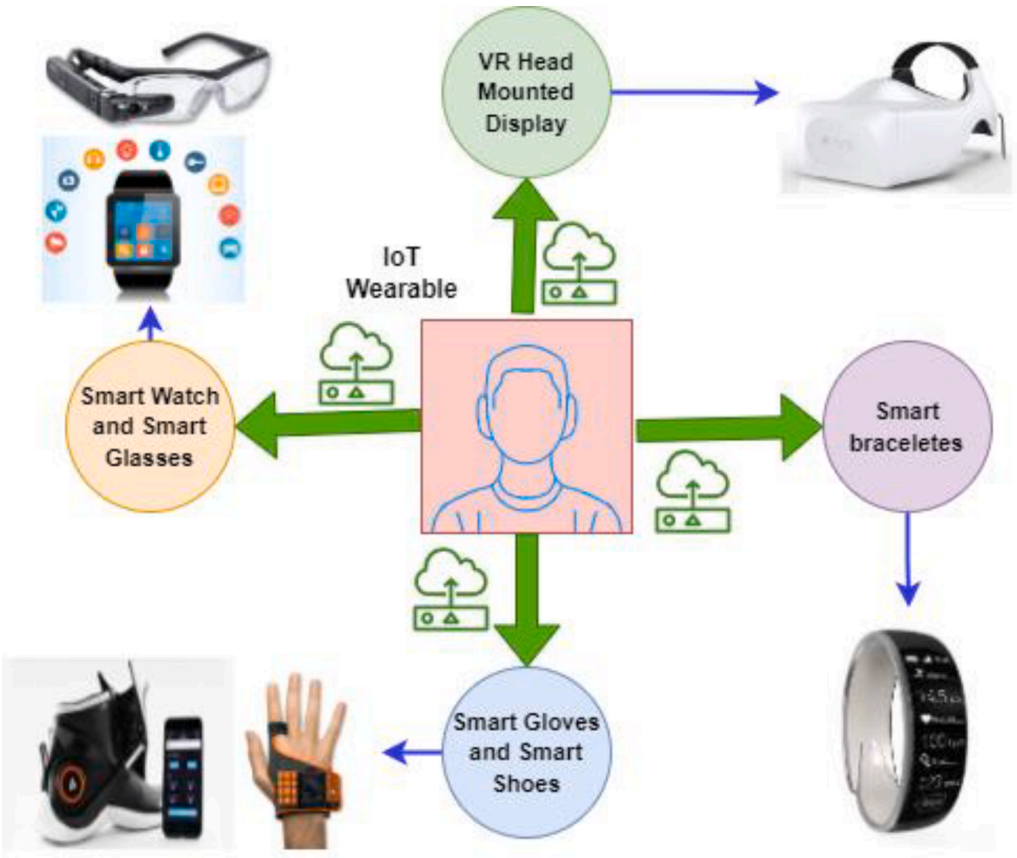


Fig. 5. Illustration of iot wearable devices.

games, smart headsets can allow players to communicate with each other during the game to help them collaborate and cooperate better.

Smart wearables sales in 2020 by Chinese companies are given in Fig. 6.

5.4. Application prospects

With the continuous development of IoT and wearable device technology, IoT wearable devices for social games will have broad application prospects.

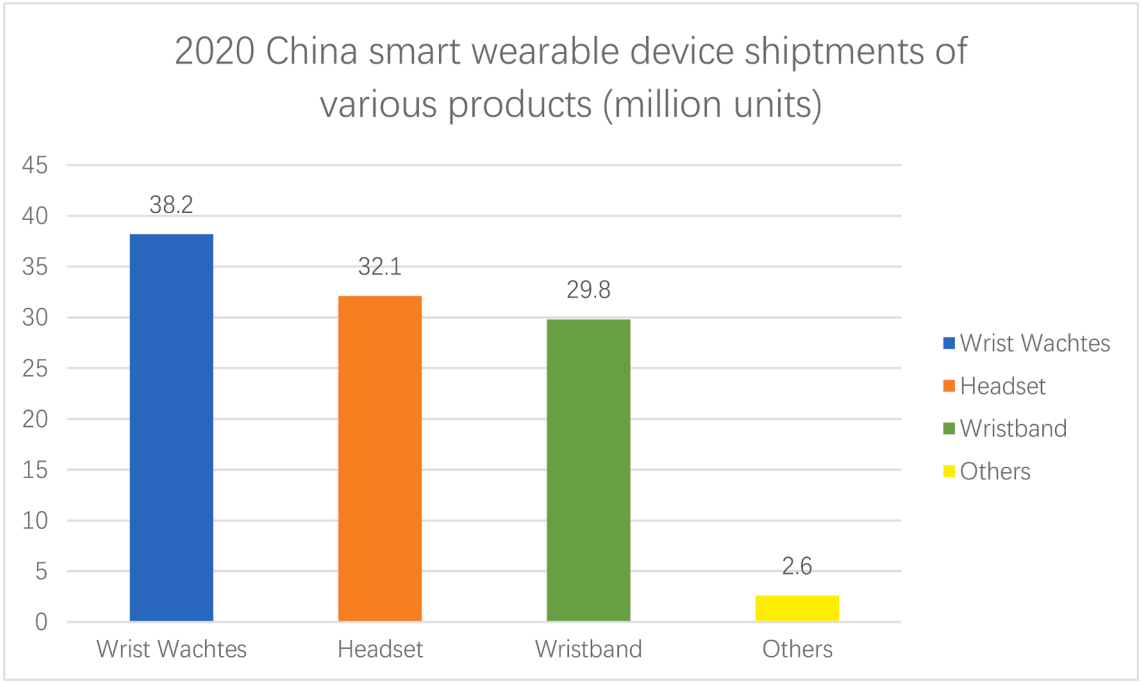


Fig. 6. Smart wearables sales of China.

Enhancement of game experience: Through wearable devices, players can experience social games more deeply, such as feeling the sound, light, vibration, etc., in the game to enhance the realism and sense of substitution of the game [74].

Enhancement of social interaction: Wearable devices can allow players to participate in social interactions in games more naturally, such as using gestures, voice, etc., to interact with other players, enhancing the sociability and fun of games [75].

Diversified application scenarios: In addition to traditional electronic games, wearable devices can also be applied to other types of games, such as somatosensory games, virtual reality games, etc., bringing users a brand-new gaming experience [76].

Application of health management: Through wearable devices, game manufacturers can collect players' health data, such as heart rate, steps, sleep quality, etc., to provide players with personalized health management services and enhance the game's sense of social responsibility and user loyalty [77].

Personalized design and customized service: Wearable devices can be designed and customized according to the user's physical characteristics and preferences, providing players with a more customized gaming and user experience enhancing brand influence and market competitiveness [56].

6. IoT wearables and sensors for serious gaming

6.1. Serious game

Serious play is a form of play designed to communicate an important message, concept, or theme. Different from traditional entertainment games, the main purpose of serious games is to convey certain education, training, publicity, inspiration, and other purposes through game experience so as to achieve a deeper learning and understanding effect [78].

Serious games often deal with real-world issues such as healthcare, social issues, environmental issues, historical events, etc [79,80]. These games can take different forms, including role-playing, simulation, interactive fiction, and more. The purpose of the game is to allow players to accumulate knowledge and experience during the game to understand better and solve problems.

Serious games can be applied in different fields, such as education, training, healthcare, politics, etc., to improve the cognitive level and practical ability of the participants.

6.2. Features of serious games

i. **Strong interactivity:** Serious games are usually highly interactive. Players can play different roles in the game and affect the game process and ending through decisions and actions [81].
ii. **Strong constraints:** Serious games usually set a series of rules and restrictions, and players need to play games under these constraints, which can simulate various limitations and challenges in real life [82].
iii. **Real-time feedback:** Serious games usually provide a real-time feedback mechanism [83]. Players can use in-game data and feedback to understand whether their performance and decisions are correct.
iv. **Strong authenticity:** Serious games usually involve problems in the real world, and their backgrounds and situations are based on real events and environments, which can give players a deeper understanding of the nature of the problem and its solution [84].
v. **Software and hardware support:** Serious games need to be equipped with corresponding hardware and software devices, such as virtual reality technology, smartphones, tablets, etc., to provide a gaming experience [85].

6.3. The development of serious games

1950 s to 1960 s: Some of the first serious games, such as simulation games and war games, appeared, mainly for training and simulation in the military and commercial fields [86].

1970 s to 1980 s: The application fields of serious games gradually expanded, including healthcare, education, urban planning, and other fields [87]. At the same time, the development of computer technology has also provided more possibilities for the production of serious games.

1990 s to 2000 s: With the popularity of the Internet and mobile devices, serious games began to develop into online and mobile platforms, such as virtual worlds, online cooperative games, etc [88].

From 2010 to now, Serious games have received more and more attention and applications, and the fields involved have become more and more extensive, such as human resource management, psychotherapy, environmental protection, public safety, etc [89]. At the same time, with the development of new technologies such as artificial intelligence and virtual reality, the form of serious games is constantly changing and innovating.

According to a report released by the market research organization Reports and Data, as of 2021, the global serious game market has reached 3.67 billion US dollars, and it is expected to reach 10.46 billion US dollars by 2028 [90]. This shows that serious games have broad application prospects in education, training, health management, and other fields. At the same time, according to incomplete statistics, more than 2,000 companies and organizations around the world have developed and provided serious game products and services, covering multiple industries and fields.

6.4. IoT sensing devices based on serious games

IoT sensing devices based on serious games can be applied in multiple fields, such as health management, environmental monitoring, industrial control, etc.

Health management: combine various health monitoring devices with games and encourage users to participate in health management more actively through games. For example, heart rate sensors, blood pressure monitors, blood glucose meters, and other devices can be connected to the game to monitor the user's health data in real time and feedback on the data to the game, allowing users to understand their health status more intuitively, and at the same time promote the user through the game. Take a more active role in managing your health.

Environmental monitoring: combine various ecological sensors with games and encourage users to participate in environmental protection more actively through games. For example, air quality sensors, noise sensors, temperature sensors, and other devices can be connected to the game to monitor environmental data in real time and feed the data back to the game, allowing users to understand the environmental conditions more intuitively and at the same time promote users to be more active through games protect the environment [91].

Industrial control: Combine various industrial sensors with games and use games to encourage employees to participate in work more actively. For example, temperature sensors, humidity sensors, pressure sensors, and other devices can be connected to the game, real-time monitoring of production data, and the data can be fed back to the game so that employees can understand the production status more intuitively, and at the same time promote employees to participate more actively through the game work and increase productivity [92].

IoT sensing devices based on serious games can combine sensor data with games to provide users with a more intuitive and personalized experience. They can also encourage users to participate more actively in activities such as health management, environmental protection, and industrial production.

7. IoT applications for gaming

7.1. IoT application

In the IoT, various devices and sensors can interact with each other, collect, transmit, and analyze data so as to realize automatic and intelligent control and management [93]. The following are the applications of IoT:

i. Smart home: IoT technology can connect various devices in the home, such as smart door locks, smart lamps, smart sockets, etc. Users can remotely control home devices through smartphones or other devices to realize automatic control and management. Improve home safety and convenience [94].
ii. Industrial Internet of Things: The Industrial Internet of Things is the digital transformation of traditional manufacturing plants through IoT technology to realize intelligent production lines and industrial automation. By connecting sensors and devices to the network, real-time monitoring and control of the production process can be achieved, improving production efficiency and quality [95].
iii. Intelligent transportation: IoT technology can realize intelligent traffic management, such as smart traffic lights, intelligent vehicle monitoring, intelligent road condition prediction, etc. By connecting traffic equipment and vehicles to the network, real-time monitoring and control of traffic data can be realized to improve traffic safety and efficiency [96].
iv. Smart medical care: IoT technology can realize smart medical management, such as smart health monitoring, smart medical equipment, etc. By connecting medical devices and sensors to the network, real-time monitoring and control of patients' physical conditions can be achieved, improving medical efficiency and accuracy [97].
v. Smart agriculture: IoT technology can realize smart agricultural management, such as smart irrigation systems, smart weather monitoring, etc. By connecting sensors and devices to the network, real-time monitoring and control of the agricultural production environment can be realized, and the efficiency and quality of agricultural production can be improved [98].

7.2. The relationship between gaming and the Internet of Things

There is a close relationship between games and the IoT, which is mainly manifested in the following aspects:

i. **Intellectualization of game hardware devices:** With the development of IoT technology, the degree of intelligence of game devices is getting higher and higher. For example, game consoles, smartphones, computers, and other devices can be connected to the network, sensors, and other technologies to achieve more Intelligent operation and interaction [99].
ii. **Enhancement of game experience:** Through IoT technology, games can be more closely connected with the real world. For example, virtual reality technology in games can bring players into a more realistic game world. IoT technology can also make games. The virtual elements in the game interact more closely with the real world, improving the player's sense of immersion and gaming experience [100].
iii. **Application of game data:** Through the IoT technology, games can collect player data, including game behavior, game preferences, and other information. These data can be used for game development, improvement, marketing, etc., to improve the user experience and commercial value [101].
iv. **Cross-application of games and IoT:** In addition to games themselves, games and IoT technologies can also be cross-applied in other fields, such as promoting IoT products and services through gamification or enhancing games through IoT technologies development and operational efficiency, etc [102].

7.3. Application of gaming IoT

Gaming IoT is the field of applying IoT technology to the gaming industry. It can provide gamers and game developers with a better gaming experience and a more efficient way to develop games.

i. Game controller: A game controller is a device used to control video games, usually consisting of a gamepad, keyboard, mouse, or other input device. Game controllers are commonly used to control the movement, manipulation, and interaction of characters or other objects in games. They can be connected to a game console or computer so that players can use them to manipulate in-game objects and environments. Different types of game controllers have different features and designs to suit different types of games and player preferences [103].
ii. Gaming devices: IoT gaming devices refer to gaming devices that are connected to the Internet and can communicate and interact with other devices or players via the Internet. These devices typically include game consoles, smartphones, tablets, TVs, and smart home devices, among others [104]. IoT gaming devices can communicate with other devices or servers through the network to realize functions such as multiplayer games, online games, and social games. In addition, IoT technology can also connect real-world objects and environments with the game world, thereby creating a richer gaming experience. For example, through the use of sensors and IoT technologies, real-world movements, sounds, and visuals can be translated into virtual elements in games in real-time, allowing players to engage more deeply in the game.
iii. Multiplayer games: IoT-based multiplayer games refer to multiplayer game modes implemented using IoT technology, allowing players to play games with other players over the Internet [105]. This game mode allows players to play games with other players anytime, anywhere, without being limited by geographical location and time, and at the same time, can achieve a richer game experience and interaction. IoT-based multiplayer games usually require players to use devices connected to the Internet, such as smartphones, tablets, or game consoles, to communicate and interact with other players via the Internet. These games can be different types of games, such as online games, social games, cooperative games, or competitive games. The advantage of multiplayer games based on the IoT is that they can achieve a more flexible and convenient gaming experience. Players can play games with other players anytime and anywhere without gathering in the same place. In addition, through the application of IoT technology, this game mode can also achieve a richer game experience and interaction, such as connecting objects and environments in the real world with the game world, making the game experience more real and immersive.
iv. Game data analysis: IoT game data analysis refers to the use of IoT technology and data analysis technology to collect, process, analyze, and mine various data involved in games so as to provide useful information for game developers and operators [106]. Information and insight. These data include player behavior data, game performance data, device data, geographic location data, etc. Through the analysis of these data, it can help game developers understand the player's behavior habits, game experience, game difficulty, game duration, etc., and then optimize and improve the game. At the same time, it can also provide game operators with useful marketing information and business decision-making basis, such as personalized marketing strategies for different player groups, prediction and optimization of game revenue, etc. IoT game data analysis plays an important role in

the modern game industry. It can help game developers and operators better understand and meet player needs and improve game quality and commercial value. Advantages of IoT in gaming are in given in Fig. 7.

7.4. Case studies of the applications of IoT in gaming

The research work presented by Balducci et al. focuses on developing and implementing an IoT-based interactive game called Magic Torch, designed to improve museum user experiences [118]. The Magic Torch game transforms the traditional museum visit into an interactive treasure hunt, engaging visitors in a dynamic learning process. The game scenario is built around using a smart device, the Magic Torch, which interacts with RFID-tagged artefacts within the museum. Visitors are divided into small groups and provided with the Magic Torch and a deck of smart cards containing questions related to the exhibits. The aim is to answer these questions by interacting with the RFID-tagged artefacts, which provide immediate feedback through the Magic Torch's audio-visual cues, such as a green light and a sound for correct answers and a red light for incorrect ones. This feedback mechanism engages the interaction and reinforces the educational content shared by the museum guide at the beginning of the visit.

The integration of IoT in gaming, as demonstrated by the Magic Torch, leverages various IoT components, including RFID readers, sensors, and actuators, connected via a NodeMCU ESP8266 microcontroller. This setup enables seamless interaction between the smart device and the physical artefacts, transforming them into interactive learning tools. The design of the Magic Torch also considers practical constraints, such as avoiding the need for proximity searches, which could cause logistical issues in a crowded museum environment. Instead, the game utilizes RFID technology to ensure reliable interaction without requiring line-of-sight or specific proximity, thus enhancing the overall visitor experience. The iterative design process of the Magic Torch involved extensive testing and feedback sessions with museum curators and guides to ensure that the game was both educational and enjoyable.

The study to evaluate the Magic Torch game involved 30 participants who experienced a guided tour of the Natural Sciences Museum at the University of Bari, followed by the interactive game. The participants' experiences were assessed using the NASA-TLX questionnaire to measure cognitive workload and the Self-Assessment Manikin (SAM) questionnaire to gauge their emotional responses. This thorough evaluation aimed to measure the game's impact on visitor engagement, learning outcomes, and overall satisfaction. The results indicated that the interactive game significantly enhanced the visitor experience by making learning more engaging and memorable.

The research paper highlights the potential of IoT applications in gaming, particularly in the context of cultural heritage sites. By leveraging IoT technology, the Magic Torch game transforms static museum exhibits into dynamic, interactive learning experiences that actively engage visitors. This approach enhances visitor satisfaction and improves educational outcomes by making learning more interactive and enjoyable. The study underscores the importance of incorporating IoT in gaming to create immersive and educational experiences, setting a precedent for future applications in various domains beyond cultural heritage. The successful implementation and positive feedback from the Magic Torch game demonstrate the effectiveness of IoT-supported interactive games in enhancing user experiences and engagement in cultural sites.

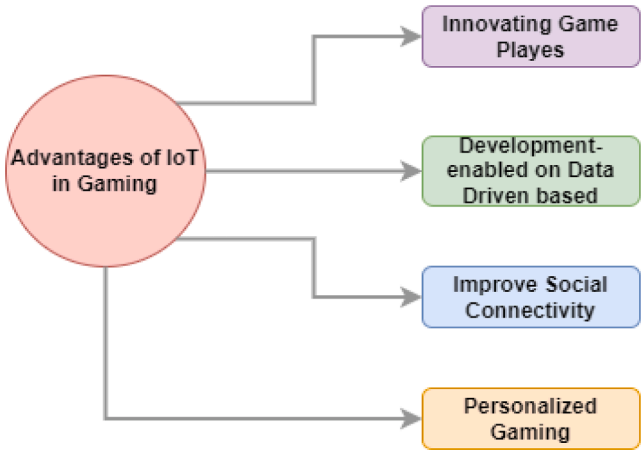


Fig. 7. Advantages of IoT in Gaming.

The IoT plays an important role in user experience and cloud gaming performance. It can enhance the connectivity of user devices, reducing latency, and enabling real-time monitoring of users' devices. IoT devices such as routers, smart gadgets, and sensors can be helpful for network infrastructure and traffic management. These devices can help players play smooth games and improve gaming data. IoT devices can also perform better in real-time applications because they can adjust the dynamic network and minimize the lag. IoT devices can be integrated into cloud gaming, home appliances, and any electric gadgets, minimizing power consumption. Another important factor is, that integration of the IoT devices into cloud gaming can protect from cyber-attacks. Furthermore, IoT devices have the facility to integrate augmented reality and virtual reality to give a better user experience in a virtual world because it can tracked through human eye. IoT devices can improve cloud gaming performance because they can be controlled from a long distance. The author investigated IoT-based games to determine how certain UX factors can influence visitors [119]. In this research, 18 young participants took part. The study was divided into two parts. In the first phase, participants were instructed and guided to visit the Museum of Natural Sciences of the University of Bari. Following this, they were asked to play the Magic Torch game and move freely within the museum according to the game's instructions. The focus was on the visitors' emotions and other aspects of their overall experience. By guiding them through both a manual and a game-based approach, the researchers could perform a quantitative analysis of the visitors' behavior during the game. According to the author, UX refers to the experience of visitors using IT products, systems, or services. This study analyzed the visitors' experience while playing the IoT-based game, as well as their experience and performance during a guided tour without using technology. The experiments showed that technology plays an important role; it was evident that emotions can be influenced by the use of technology. When visitors played the game using IoT devices, their results improved, and they performed better. They also found that the workload of visitors as compared to traditional tour guides is not affected negatively by visitor's emotions and their experience. They have received positive feedback regarding their experiments. The author has identified that the integration of IoT in games will give positive feedback.

8. Challenges and limitations

The challenges and limitations developers and designers face in IoT gaming are given in Fig 8. and described in detail in subsections.

8.1. Technical challenges and limitations

i. Network security issues: IoT devices are connected to the Internet and communicate with other devices and systems, which makes IoT systems face the risk of cyber-attacks and data leakage [107]. In this case, ensuring the security of the IoT system is of paramount importance.
ii. Data management and processing issues: IoT devices usually generate a large amount of data; effectively managing and processing this data is an important challenge. IoT systems need to be

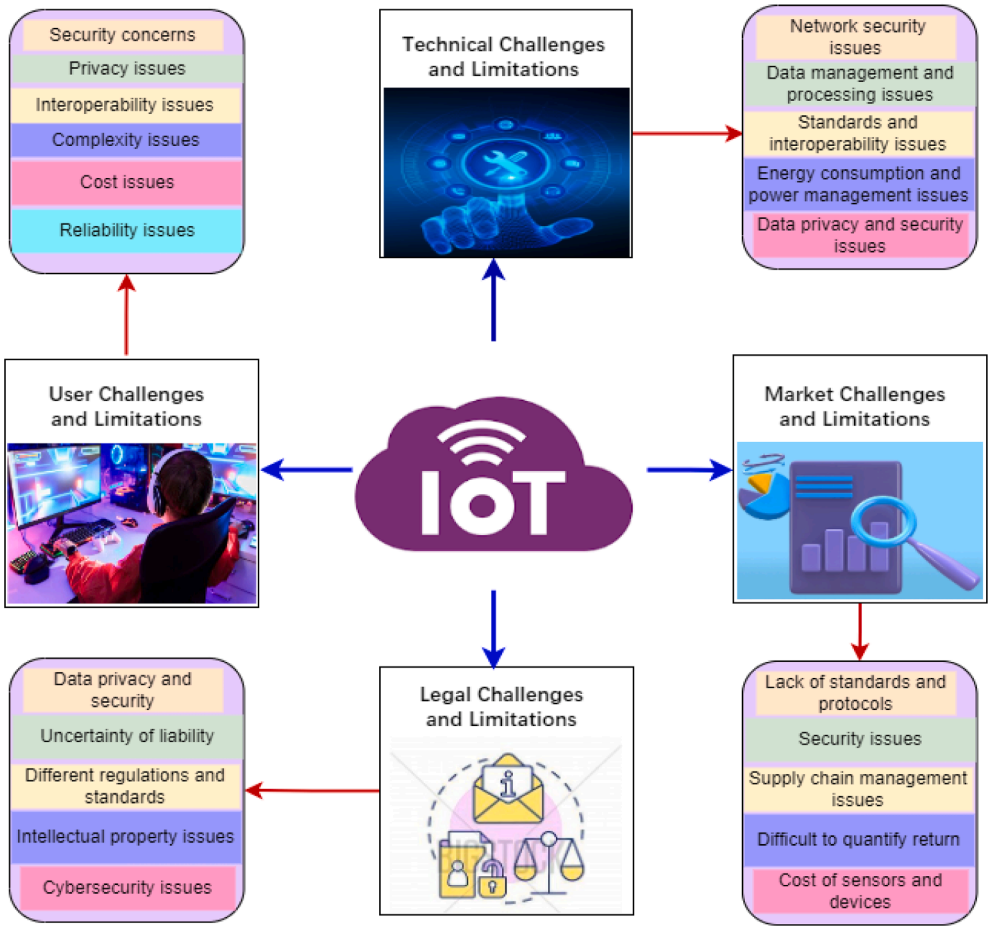


Fig. 8. Challenges and limitation of iot in gaming.

able to collect, store, process, and analyze this data and extract useful information from it [108].

iii. Standards and interoperability issues: Since IoT technologies involve multiple technology domains and different suppliers, different standards and protocols may exist [109]. This may lead to incompatibility between other devices, thereby limiting the application range and market size of IoT technology.

iv. Energy consumption and power management issues: IoT devices often need to run for a long time, but the battery life they use may be limited. Therefore, energy consumption and power management are important issues for IoT technology, and it is necessary to find effective ways to extend the battery life of IoT devices [110].

v. Data privacy and security issues: IoT devices usually involve a large amount of sensitive data, such as personal health records, financial information, etc.

8.2. Market challenges and constraints

i. Lack of standards and protocols: Since IoT technology involves multiple technical fields, different suppliers and devices may use different standards and protocols. This may lead to incompatibility between different devices, thereby limiting the application range and market size of IoT technology.

ii. Security issues: The security issues of IoT devices and systems have become an important challenge. Since IoT devices usually involve a large amount of sensitive data, such as personal health records, financial information, etc., security breaches may lead to serious privacy leaks and financial losses. Therefore, ensuring the security of IoT systems has become an important factor in marketing.

iii. Supply chain management issues: IoT systems are often built from multiple suppliers and components, so supply chain management becomes an important issue [111]. If there is a problem with one of the suppliers, the reliability and stability of the entire system may be affected.

iv. Difficult to quantify return: While IoT technology has many potential business uses, it may be difficult for businesses to quantify the return of IoT technology because the technology is relatively new. This can lead to under-investment in IoT technology by businesses, restraining the growth of the IoT market.

v. Cost of sensors and devices: IoT systems involve a large number of sensors and devices, and the cost of these devices may limit the application scope and market size of IoT technologies [112].

8.3. Legal challenges and limitations

a. Data privacy and security: With the popularity of IoT devices, a large amount of personal data will be collected, stored, and transmitted. This data may contain sensitive information such as medical records, financial information, and personal location data. Therefore, data privacy and security issues become an important challenge. The government and the industry need to formulate relevant regulations and standards to protect user privacy and security.

b. Uncertainty of liability: IoT devices and systems are built from multiple components and suppliers, which can lead to uncertainty of liability. For example, if a self-driving car crashes, who should be responsible? Manufacturer, software developer, or vehicle owner?

Accordingly, a corresponding legal framework needs to be developed to address liability issues.

c. Different regulations and standards: Due to the rapid development of Internet of Things technology, perfect regulations and standards are needed to regulate its development and application. There are also differences in rules and standards in different countries and regions, which may affect the global application of IoT technology.
d. Intellectual property issues: IoT devices and systems involve multiple technical fields, such as communications, computer science, electronic engineering, etc. This can lead to intellectual property issues such as patent infringement etc. Therefore, it is necessary to formulate corresponding intellectual property laws and regulations to protect the interests of relevant parties.
e. Cybersecurity issues: IoT devices and systems are part of the Internet and, therefore, also face cybersecurity threats such as hacking and malware. These security issues may lead to serious consequences such as system failure and data leakage. Consequently, it is necessary to formulate corresponding network security regulations and standards to ensure the security of IoT systems.

8.4. Challenges and limitations in terms of user needs

a. Security concerns: IoT devices often need to be connected to the internet, which makes them vulnerable to hacking. Hackers can use loopholes, password weaknesses, etc., to invade IoT devices and then obtain user data, control devices, and launch network attacks. IoT devices are at a higher risk of potentially being attacked than traditional devices because they typically use open-source software and standard protocols, meaning hackers can more easily obtain information about them.
b. Privacy issues: IoT devices can collect a large amount of user data, including location, health status, social activities, and more. This data may contain personal information such as name, address, phone number, etc. If this data is stolen or misused, the user's privacy may be violated. In addition, IoT devices may also share data without the user's knowledge, which may lead to insecurity and misuse of user data.
c. Interoperability issues: Since IoT devices come from different vendors, they may use different protocols and standards. This can cause interoperability issues between devices. For example, one smart home device may not be able to communicate with another smart home device because they use different protocols. This means that the user may need to purchase matching equipment to achieve the required functions, which increases the cost and complexity for the user.
d. Complexity issues: IoT devices usually have complex functions and configurations, and users may need to spend a certain amount of time and effort learning and using these devices. For example, a smart home system might require the user to learn how to use a phone app, set rules and scenarios, etc. This may cause some distress for older people and the less technologically literate.
e. Cost issues: IoT devices are usually more expensive than traditional devices. This is because IoT devices usually require more advanced technologies, such as sensors, cloud computing, and big data analytics. In addition, IoT devices require higher security features, which also increases their cost.
f. Reliability issues: IoT devices may malfunction or become unavailable, which may prevent users from using desired functionality. For example, a smart home system might not be able to control lights or temperature because a device malfunctions or a cloud server fails to connect [113]. This may cause some inconvenience to the user's life.

9. Open research issues

IoT technologies, such as wearable devices, emerged in gaming, but network security for data transfer is still a problem for game players [114]. If someone enters the command for the next step of the game, that hacker will change, decreasing the user's enjoyment level. Developing more secure IoT gaming apps and platforms will attract more organizations and game players, so future research requires more secure IoT gaming. IoT still lacks standards and protocols, so developing robust gaming standards will improve performance [115]. It is challenging to configure IoT gaming wearables with mobile phone apps, so in the future, tutorials and guidelines for using and connecting with other apps will be helpful for users [116]. In the future, the development of apps that automatically configure devices with apps will make it easier for users.

Different vendors set different standards and protocols for IoT devices, which is also a big problem for users and organizations configuring and managing IoT infrastructure for gaming platforms [117]. Industry developers need to develop open standards and protocol devices of IoT that will be easily configurable with all other vendors' designed IoT hardware (sensors and actuators). The cost of the sensors and other IoT devices is still very high. Hence, there is a need to develop high-quality IoT devices for future usage, which will be purchasable for people of economic countries, increasing IoT gaming and generating revenue for service providers.

10. Conclusion

IoT plays a vital role in the design and development of games. In this paper, we reviewed the role of IoT, such as wearables sensors, in severe and social gaming. We also discussed IoT definitions, key concepts, and architecture in detail. Challenges and limitations of IoT gaming include technical, legal, market-based, and user perspectives concerning previous research. We added open research issues of IoT gaming for further development to provide solutions to organizations and users for a better gaming environment. In future work, we plan to conduct quality of experience (QoE) of IoT-based games using different wearables devices to analyze the user enjoyment/engagement with games and compare them with previously developed non-IoT games. The study will elaborate on users' needs for future game development and better utilization of IoT in games.

CRediT authorship contribution statement

Asif Ali Laghari: Writing – original draft, Methodology. **Hang Li:** Supervision, Conceptualization. **Yin Shoulin:** Software, Resources. **Awais Khan Jumani:** Methodology, Formal analysis. **Abdullah Ayub Khan:** Writing – review & editing, Resources. **Fida Hussain Dahri:** Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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